

IMPACT OF BUILDING PARAMETERS ON CONSTRUCTION COST USING ARTIFICIAL NEURAL NETWORK

Prasad H.B.

Commercial/ QS Department
Shapoorji Pallonji Mideast LLC
Dubai, United Arab Emirates
bhawantha.prasad@shapoorji.com

Gunarathna M.G.C.H.L.

Faculty of Engineering
NSBM Green University
Homagama, Sri Lanka
chamil.g@nsbm.ac.lk

Madushanka T.H.

Faculty of Engineering
NSBM Green University
Homagama, Sri Lanka
hashan.m@nsbm.ac.lk

Guruge R.N.D.

Faculty of Engineering
NSBM Green University
Homagama, Sri Lanka
romasha.g@nsbm.ac.lk

Abstract— The construction cost is influenced by numerous factors, hence is difficult to develop an accurate estimation without identifying these factors. Therefore, it is more important to examine and rank the most cost-affecting factors on construction costs. Therefore, this research aims to identify the impact of factors and parameters on the construction cost of the buildings. Under that, residential building projects with less than 4 stories have been selected as building types for this study. Further, cost estimation at the early stage is critical for successful completion within the budgets. However, it was difficult to develop an accurate estimation using traditional estimation methods. Therefore, several researchers have proposed a solution as the parametric estimation based on building parameters for that. Therefore, this study has developed a parametric cost prediction model using computer-based tools. Further, this study has suggested an elemental cost estimation because it was more accurate than other methods. Finally, the research identified the impact of building parameters on different building elements based on that prediction. Artificial Neural Network (ANN) was used as a computer-based tools for study. Initially, a comprehensive literature review was conducted to determine 21 factors and 24 building parameters affecting construction cost. Then, thirty-two questionnaires were distributed among construction professionals to identify the significant factors and building parameters for the construction cost of low-rise residential buildings. Moreover, the questionnaire survey identified input variables for each ANN model to predict the cost of building elements. Further, twelve case studies of low-rise residential building projects were selected from the construction industry in Sri Lanka. These case studies were used to train the cost prediction models for each building element. The selected parameters for the cost of building elements were considered as independent variables that affected the construction cost of the building element. The ANN model predicted the cost of building elements based on inputs and it measured the importance of inputs for the acceptable prediction. The research has identified ten factors as highly significant in construction cost based on a questionnaire survey. Moreover, nine factors were categorised as significant. Further, the ANN model shows the importance of building parameters on each building element based on the

prediction. Among them, GIFA, number of floors, height between floors and external wall area are more influential building parameters on most the building elements.

Keywords- Building construction cost, Artificial Neural Network, Cost estimation, building parameters, Sri Lanka

I. INTRODUCTION

Estimation of construction cost is one of the challenging tasks in construction management not only past but also nowadays (Adeli & Wu, 1998; Ellina, Papaschinopoulos & Papadopoulos 2019). Construction clients always require early and accurate cost advice before they commit to their investments (Lowe, Emsley & Bandara, 2006). Therefore, in the feasibility stage, the construction contract price is forecasted to assist the client in setting the budget (Ashworth & Perera, 2015). However, since information is limited at an early stage, it is more difficult to forecast and plan the cost of the project accurately (Koo, Hong & Hyun 2011). The cost of construction depends on the various factors which are affected during the whole project duration (Belniak, Leśniak, Plebankiewicz & Zima, 2013).

Artificial Neural Networks (ANNs) are an advanced computational model capable of handling complex datasets for prediction and analysis. Unlike traditional regression methods, ANNs can model non-linear relationships and learn from data patterns, making them particularly suitable for cost estimation in construction projects. This study employs ANNs to predict construction costs more accurately by leveraging historical project data and identifying key cost-driving parameters.

The accuracy of estimation is also critical for the success of the project (Kim, An, & Kang, 2004). Because, accurate estimation does not just give a figure but also advises on future decisions (Gerardi, 2021). According to Shash and Abdul-Hadi, (1992), the factors affecting the accurate

estimation were categorized as financial issues, bidding situations, building characteristics, accuracy of available information and estimation methods and techniques. Further, they have found that building characteristics and information availability are critical factors in cost deviation in construction projects. Another research related to construction cost estimation was done by Akintoye (2000) who also stated that estimation methods and available project information are the main factors that affect construction cost estimation practice at the early stages of projects. Hence, it is needs to collect correct and suitable project information for estimation. Then, suitable methods should be used to get much more accurate results.

Several researchers have used different modern techniques to analyze and predict the conceptual cost of the building. Probability analysis, artificial neural networks, fuzzy analogy method, time series forecasting and case base reasoning are some methods that used to predict the cost (Xu, Xu, Zhou, & Wu, 2015). Probability analysis methods are much more complex and they need a comprehensive database of previous projects (Dursun & Stoy, 2016). However, in the Sri Lankan context, there is no such database. Similarly, for case base-reasoning methods also need detail in a large number of projects to get accurate results (Xu et al., 2015). Fuzzy is also a prediction model for human decision-making. It is much more difficult to develop the fuzzy rules for the cost data. However, artificial neural networks have been used as successful models in some studies (to analyze and develop cost prediction models (Arafa & Alqedra, 2011; Attalla, Hegazy, & Asce, 2003; Emsley, Lowe, Duff, Harding, & Hickson, 2002; Sawalhi, 2012; Sawalhi & Shehatto, 2014). Therefore, the artificial neural network can be used as a successful analysis method to project data.

The traditional quantity surveyors of Sri Lanka often failed to estimate accurately at the conceptual stage of construction (Silva, Rajakaruna, & Bandara, 2005). Moreover, the authors found that the traditional techniques of estimation are the major challenges facing most of the construction organizations in Sri Lanka. However, parametric cost models have been developed as accurate estimation methods in several regions (Sistani & Rezaei, 2016). Further, this parametric estimation is always based on historical building data. Therefore, it gives much more accurate results. For example, 20 high-rise building projects in the United States, were analysed to demonstrate a cost model by Sonmez (2008) using parametric cost models. 16 variables were used based on the characteristics of the buildings in both the commercial and residential sectors. The most common method to estimate the cost in a conceptual stage in Sri Lanka is expert judgment. They have been using the comparison of similar projects based on their professional experiences. However, the accuracy level of this method is highly dependent on the experience of estimators (Perera, A.d., & Illankoon, 2013). Hence, the Sri Lankan construction industry needs to investigate systematic ways of cost prediction methods for the conceptual stage, to increase construction productivity

(Rathnayake, Mollasalehi, Aboumoemen, Kulatunga, & Samir, 2018).

II. LITERATURE REVIEW

A. Factors influencing construction cost

The cost, quality and time are the three main drivers to the successful completion of a construction project (Phaobunjong, 2002). These are also interdependent parameters in every construction project (Hu & He, 2014). It is a tough challenging task to balance cost, quality and time correctly (Ballhysa & Blloku, 2014). In most projects, the specific quality and time are decided depending on the available budget (Hameed Memon, Rahman, Abdullah, Asmi, & Azis, 2014). Furthermore, the authors revealed that cost is the most important driver to control both quality and time.

The factors influencing construction cost can be divided into four categories such as project-specific factors, client-contractor-related factors, competition and market conditions and macroeconomics and political factors (Sawalhi, 2012). Further, design-related factors and site-related factors are more detailed classifications of project-specific factors (Emsley et al., 2002). Moreover, these design-related factors can be divided into more groups such as size-related variables, structure-related, finishes-related variables and systems and equipment-related variables (Juszczak, Leśniak, & Zima, 2013). The project-specific factors are more influencing factors for both the total amount and the unit prices of input resources (Tas & Yaman, 2005). According to Cunningham (2013), the project-specific factors can be listed as function, design parameters, choice of materials and so on. Site condition also includes different factors like site area, access to the site, soil condition and so on (Emsley et al., 2002). The client priorities such as quality considerations, and time considerations could influence the construction cost of any kind of project (Tas & Yaman, 2005). On another hand, the nature of the client directly impacts the tender process and causes changes in construction cost (Silva et al., 2005). Competition and market also directly affect to total contract amount and unit price of elements (Cunningham, 2013). Inflation interest rate fluctuations, labour laws, construction regulation and other legislative constraints can be grouped under macroeconomics and political factors which may be highly impacted by cost and delays in construction projects (Warsame, 2006). Other than these categories, social and environmental factors such as environmental conditions and regulations, and cultural aspects, the region could influence the cost of projects (Sawalhi, 2012). Similarly, different studies have selected factors as cost variables for building construction. According to Table 1, large numbers of factors can be categorized as design-related variables. Among those factors, size-related variables have been commonly used. Secondly, site-related

variables are selected. Therefore, it has been shown that design-related variables are the most effective factors in construction costs.

Table I-1- Factors Affecting on Construction Cost of a Building

Project-related factors	Construction site-related factors
Project duration	Site location
Procurement method	Site topography
The contract type	Soil condition
Tendering strategy	Quality of workmanship
Design team experiences	Major demolition on site
Pre-qualified contractors	Site waste treatment
	Special site preparation requirements
	Site management and communication
Client related factors	Macroeconomics & political factors
Nature of the client	
Quality requirement of client	Government regulations and policies
Financial ability of the client	Construction cost index
Materials specified by client	
Market conditions	
Material, labour, plant, availability	

Design related building parameters	
Gross internal floor area	Type of frame (steel/ concrete)
External walls area	Type of roof
Internal walls area	Type of slab
Ceiling area	Type of external plastering
Roof area	Types of ceiling finishes
Basement floor area	Type of floor finishes
Common areas	Type of electrical works
Numbers of floor	Type of mechanical works
Floor height	Types of protective installations (fire & security)
Total height of the building	Volume of HVAC
Numbers of doors and windows	Numbers of column within a floor
Type of foundation	Parking areas within the building

B. Construction cost estimation

The primary function of cost estimation is to prepare a reliable cost prediction for a project (Elsaway, Hosny, & Rezek, 2011). Since the construction project uses a huge amount of capital and has a high level of risk factors, it is important to prepare a cost prediction (Ji et al., 2019). The construction estimates are being used to advise on the expenditure needed to complete the project (Rathnayake et al., 2018). Cost estimation is not just a figure but also advises the client about components, directly and indirectly, impact costs like material, labour, machinery, location, duration, building codes, construction facilities and so on (Gerardi, 2021). Further, cost estimation always facilitates more effective time scheduling, quality controlling and resource managing (Elsaway et al., 2011). On the other hand, the cost managers have to allow contingency in the project cost during the decision-making process. Consequently, accurate cost estimation is always used to support a confident decision-making process (Lee et al., 2019).

C. Factors affecting accurate construction cost estimation

Accuracy of the estimation of construction cost is crucial for project budgeting, planning, and monitoring in terms of available budget, time and outstanding works (Prechelt, 1998). As well as it will also be a critical factor for

transparent construction projects and the successful completion of the project (Rathnayake et al., 2018).

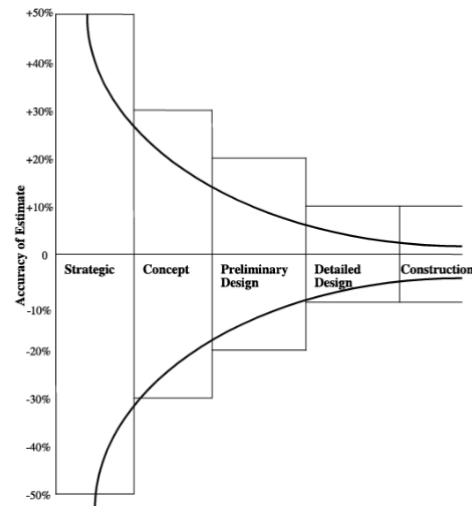


Figure II-1-The accuracy of the estimation in different stages.

Adopted from Ballard & Pennanen, (2013)

The available project information and experience level of the estimator are the most significant factors in traditional cost estimation (Sawalhi, 2012). Further, the accurate estimation will depend on the complexity of the project, technology used for estimation, experience and skill of the project team and market conditions (Akintoye, 2000). On the other hand, an inaccurate estimation can arise from two sources such as problems associated with the project and problems associated with estimating techniques and operating (Perera et al., 2013).

D. Parametric estimations

Parametric estimation is a mathematical technique to determine the cost by using collected historical data related to a particular product. Although it is needless details, the captured cost has a high level of accuracy (Sawalhi & Shehatto, 2014). Therefore, several studies have suggested this parametric method as an estimation method at the conceptual stage of the project (Oberlender & Trost, 2001). In the conceptual stage, available information and estimation methods are limited. Therefore it needs an appropriate technique to deal with this limited information and give a much accurate estimation (Sawalhi & Shehatto, 2014). The important parameters have been used to develop cost predictions using past project data (Sebastian, 2020). The parametric estimation is based on the information of building parameters (Sonmez, 2008). Therefore, questionnaire surveys, expert interviews, literature reviews and past project information are common methods of selecting various building parameters for the estimation process (Elmousalami, 2020). Then, it is commonly used regression analysis and neural network techniques to determine the parametric cost models (Lowe et al., 2006). Regression analysis is one of the common techniques used to sort out effective variables of

construction project cost (Sonmez, 2008). The neural network is also used as a successful technique same as regression analysis (Kim et al., 2004). On the other hand, the conceptual estimate is conducted with a high level of uncertainty (Sebastian, 2020). Therefore, is needed systematic justification to interpret the estimation results (Elmousalami, 2020). Parametric estimation is a method of making point estimate and it will be the best solution for the uncertainty of estimation.

E. Artificial neural networks

Artificial neural networks (ANN) are a new approach based on the theory of computer systems to analyse data sets (Elmousalami, 2020). Moreover, Computer-based analysis methods have grown popular in reason years for research analysis than manual analysis methods. ANN is a method that can deal with analysis involving incomplete data sets, or incomplete information (Sawalhi, 2012). Moreover, it is commonly used in testing and prediction methods for highly complex data analysis. Further, ANN can deal with linear and non-linear problems and give a much more accurate output. ANN is commonly used in prediction tasks because it can analyse and learn from experiences or examples and then predict future changes (Sawalhi & Shehatto, 2014). Another advantage of this ANN is the ability to understand more difficult mathematical formulas than other methods such as regression and correlation (Emsley et al., 2002). Moreover, this kind of computer-based analysis method can produce outputs based on different inputs and without presenting more complex mathematical functions.

Several researchers have applied neural network models in various tasks in construction sectors. Cost prediction and cost impact analysis in different stages are major tasks among them. In Sri Lanka, a limited number of research conducted based on artificial neural network approaches. However internationally, Arafa and Alqedra (2011) developed a cost prediction model based on ANN to use at an early stage of the project. It was a successful model in predicting the construction cost. However, it only included basic information about buildings. In the same year, Elsaway et al. (2011) have used ANN to predict the site overhead cost for building projects in Egypt. Sawalhi & Shehatto, (2014), have also developed a cost model based on ANN for building projects. It also included 12 input variables with one hidden layer to predict the total construction cost.

However, there are limitations in cost elemental cost analysis using such kind of model. It will be more accurate than previously developed methods because the parameters and cost details used in the model are more specific and the number of inputs can be reduced. Further, if the input variables are reduced, the accuracy of the ANN model with one hidden layer is high. Therefore, elemental cost analysis will be more suitable for the early stages of projects. Moreover, less sample data can also be used for a limited number of variables.

SPSS is the software that would be used to develop the ANN models. The following steps can be used to implement a neural network model using SPSS.



Figure II-2- Process of implementing ANN mode.

Adopted from; Sawalhi & Shehatto, (2014)

III. RESEARCH METHODOLOGY

The quantitative research approach was conducted to achieve the aim and objectives of the research. The quantitative approach gives flexible techniques to develop a systematic analysis. Yin (2011) has described that a quantitative approach is suitable if the researcher investigates existing phenomena, current situations, opinions, aspects, and in-depth study on topics with numerical interpretation. First, the comprehensive literature review is carried out to identify and understand the research gap and practices of factors affecting construction cost, and different cost estimation methods in the conceptual stage. The survey strategy and case study research strategy were selected as research strategies. The survey strategy is centred on different sizes of groups of individuals by choosing and analysing on sample chosen from these groups. The fundamental reason for this strategy is to find the relative frequency, conveyance and interrelationship of factors (Tamura et al., 2011). A questionnaire survey was used as a data collection technique for the survey. The data from the case study can be collected through interviews, questionnaires, reports, and archival records. Further, the cases should be selected to collect maximum information for the study. For this research, low-rise residential building projects were selected to collect data related to building parameters and the cost of building elements. These data were collected based on the documents of drawings and BOQs. To analyse the results of the questionnaire survey to determine the significant cost impact factors, the RII analysis method was used. The following formula was used to calculate the RII value of the factors.

$$RII = \frac{\sum (W \times n)}{(A \times N)}$$

Where,

W = Constant expressing the weighting to each response

A = Highest weighting

n = frequency of responses

N= Total number in the responses

According to the Hiyassat et al. (2016), four scale levels are transformed from RII values and the Table was derived from those intervals.

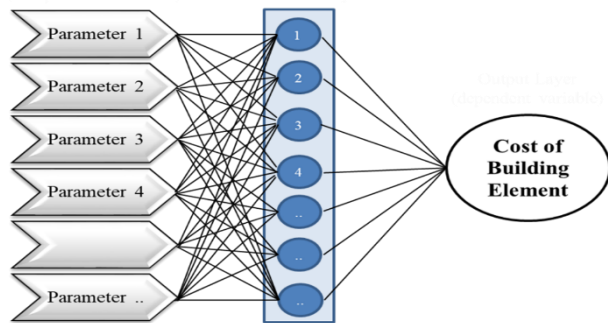
RII value level criteria

RII Value	Value level
$0.80 \leq \text{RII value}$	Highly significant impact
$0.60 \leq \text{RII value} < 0.80$	Significant impact
$0.60 > \text{RII value}$	No significant impact

Adopted from Hiyassat et al. (2016)

Based on the output of the questionnaire survey, the input parameters were selected for the model. Then, project details of 12 low-rise residential projects were collected to develop the ANN model. Initially, all collected data were extracted data to an Excel spreadsheet. Input data of building parameters and cost details of selected projects were organized before encoding in SPSS. ANN only deals with numeric data inputs. Hence, it needed to convert most of the raw data to the numeric forms. Therefore, as the first step textual inputs were defined as numerical value. Then, all numerical data is encoded in SPSS sheets. These encoded data included both details of building parameters and the cost of each building element.

The artificial neural network includes three layers an input layer, a hidden layer and an output layer. The input layer was used to enter the details of building parameters as independent variables for the model. This study developed ANN models for each building element. Input variables for each model were selected by questionnaire survey. The output layer is the layer that shows the results of the study.



The dependent variable is shown in this layer. For this study, the cost of each building element is the dependent variable and cost prediction is done for it. The hidden layer is the process layer to identify the relationship between the input layer and the output layer. Deciding the number of hidden layers is another task in ANNs. However, there is no method or rule to determine the number of hidden layers for analysis. Sawalhi & Shehatto, (2014) used that one hidden layer with 11 input variables. Further, it had also been recommended to use one hidden layer for less than this number of inputs.

According to the SPSS neural network, there are two types of neural networks such as Multilayer Perception (MLP) and Radial basis function. Among them, Sawalhi & Shehatto, (2014) has stated that MLP is a powerful tool for future prediction. Therefore, MLP was used for this study. Further, this model includes two main actions such as training the models and testing the model. During this training process, randomly selected data are available data to train the model. Then the testing phase of the model is used to ensure that the model was trained successfully. These two phases are critical to minimize network error. Then, the ANN model predicts the cost and shows the importance of each input parameter in prediction.

IV. RESEARCH FINDINGS

The research identified ten highly significant factors and nine significant factors affecting construction costs based on a survey. The ANN model highlighted the importance of specific building parameters for cost prediction. Key influential parameters include Gross Internal Floor Area (GIFA), number of floors, floor height, and external wall area. These findings underscore the critical role of certain building parameters in cost estimation, providing valuable insights for improving accuracy in the construction industry.

A. Most significant building parameters on construction cost of low-rise residential building

The parameters were sorted according to the RII significance level of cost impact. The impact of all project-related factors, client-related factors, construction site-related factors, market conditions and macroeconomics & political factors were measured by considering the total construction cost of buildings. The table shows these factors along with RII significance the level of cost impact, Means and ranking of building parameters on construction cost.

Table IV-1- Level of impact of factors on construction cost.

Rank	Factors	RII - Significance level of Impact	Mean	Significant level
1	Material, labour, plant, availability	0.953	3.813	Highly Significant
2	Project duration	0.938	3.750	Highly significant
3	Materials specified by client	0.922	3.688	Highly significant
4	Design team experiences	0.906	3.625	Highly significant
5	Quality of workmanship	0.891	3.563	Highly significant
6	Nature of the client	0.883	3.531	Highly significant
7	Quality requirement of client	0.883	3.531	Highly significant
8	Site location	0.875	3.500	Highly significant
9	Financial ability of the client	0.875	3.500	Highly significant
10	The contract type	0.813	3.250	Highly significant
11	Pre-qualified contractors	0.789	3.156	Significant
12	Site topography	0.773	3.094	Significant
13	Soil condition	0.750	3.000	Significant
14	Procurement method	0.695	2.781	Significant
15	Tendering strategy	0.656	2.625	Significant
16	Government regulations and policies	0.656	2.625	Significant
17	Major demolition on site	0.625	2.500	Significant
18	Special site preparation requirements	0.617	2.469	Significant
19	Site management and communication	0.609	2.438	Significant
20	Site waste treatment	0.531	2.125	No significant

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B. Significance of building parameters on construction cost of different building elements

According to the BCIS (2012), the substructure is the first main element of the building. The substructure cost includes the cost of the foundation and other supporting structures. The identified parameters from the questionnaire survey were used as independent variable inputs for the model and the dependent variable was the cost of the substructure. Then, the ANN indicated the importance of each independent variable against the dependence variable of substructure cost.

Table IV-2-Importance of building parameters on cost of substructure construction

	Importance	Normalized Importance	Rank
GIFA	0.414	100.0%	1
External wall area	0.251	60.6%	2
Floor height	0.138	33.2%	3
Total height of the building	0.129	31.1%	4
Type of frame	0.046	11.2%	5
Type of foundation	0.013	3.3%	6
Numbers of floors	0.009	2.2%	7

C. Impact on construction cost of super-structure frame

Four building parameters were identified as having a highly significant impact on this frame construction which have an RII value range between 0.8 and 1.0 as per the questionnaire survey. Moreover, two parameters were identified as significantly impacted parameters. ANN is predicting the cost, and the results show a good linear relationship between actual and predicted cost. Based on that, the following table shows the ranking of parameters-based importance of each building parameter on the model.

Table IV-3-Importance of building parameters on construction cost of superstructure frame

	Importance	Normalized Importance	Rank
GIFA	0.666	100.0%	1
Floor height	0.112	16.8%	2
Total height of the building	0.097	14.5%	3
Numbers of floors	0.069	10.3%	4
Type of frame	0.056	8.3%	5

D. Impact on construction cost of super-structure upper floors and stairs

The next building element is the superstructure upper floors and stairs. The questionnaire survey identified the design-related building parameters that are significant in the construction cost of this element. Similarly, the ANN has indicated the importance of each independent variable.

Table IV-4-Importance of building parameters on construction cost of super-structure upper floors and stairs

	Importance	Normalized Importance	Rank
Gross internal floor area (GIFA)	0.678	100.0%	1
Floor height	0.180	26.5%	2
Number of floors	0.142	20.9%	3

E. Impact on the construction cost of super-structure upper floors and stairs

According to BCIS (2012), external walls and internal walls or partitions are categorized into this element. Initially, the questionnaire survey distinguished factors that significantly impacted and not significant factors on the construction cost of a particular element in low-rise residential buildings. The importance of each parameter for this prediction is also indicated by the ANN model.

Table IV-5- Importance of building parameters on the construction cost of super-structure wall including external walls & internal walls/partitions.

	Importance	Normalized Importance	Rank
External wall area	0.291	100.0%	1
Numbers of floors	0.203	70.0%	2
Internal wall area	0.189	65.0%	3
Gross Internal Floor area (GIFA)	0.116	39.7%	4
Floor height	0.057	19.8%	5
Numbers of windows	0.053	18.2%	6
Numbers of Doors	0.046	15.7%	7
Total height of the building	0.046	15.7%	8

F. Impact on the construction cost of roof

The roof is the next main building element that was selected as per the BCIS (2012). Moreover, the roof includes roof decks or slabs, all structural elements of the roof, roof boarding, gutters, downpipes, necessary plates and so on.

Table IV-6- Importance of building parameters on construction cost of roof

	Importance	Normalized Importance	Rank
Roof area	0.410	100.0%	1
GIFA	0.373	91.1%	2
Type of roof	0.277	67.7%	3
Total height of the building	0.073	17.7%	4
Numbers of floors	0.067	16.4%	5

G. Impact on construction cost internal/external doors and windows

The next building elements are internal and external doors and windows. Initially, the questionnaire survey distinguished parameters that significantly impacted and not significant factors on the construction cost of a particular element.

Table IV-7- Importance of building parameters on construction cost internal/external Doors and windows.

	Importance	Normalized Importance	Rank
Numbers of doors	0.262	100.0%	1
Number of windows	0.236	90.1%	2
Numbers of floors	0.202	77.0%	3
Internal wall area	0.080	30.4%	4
Floor height	0.079	30.3%	5
GIFA	0.073	27.9%	6
External wall area	0.068	25.9%	7

H. Impact on construction cost of wall finishes, floor finishes and ceiling finishes

According to the BCIS (2012), finishes include wall finishes, floor finishes and ceiling finishes. This section discussed wall finishes including external and internal wall finishes. Then, the ANN model ranked the parameters based on the cost prediction.

Table IV-8- Importance of building parameters on the construction cost of wall finishes

	Importance	Normalized Importance	Rank
Internal wall area	0.204	100.0%	1
Type of external plastering	0.181	88.9%	2
External wall area	0.150	73.3%	3
Numbers of Doors	0.147	72.0%	4
Numbers of windows	0.141	68.9%	5
Numbers of floors	0.085	41.8%	6
Total height of the building	0.044	21.8%	7
Floor height	0.030	14.7%	8
GIFA	0.018	8.6%	9

Table IV-9- Importance of building parameters on construction cost of floor finishes

	Importance	Normalized Importance	Rank
GIFA	0.438	100.0%	1
Number of floors	0.412	94.2%	2
Type of floor finishes	0.084	19.1%	3
Internal wall area	0.038	8.7%	4
External wall area	0.029	6.6%	5

The prediction of the construction cost of ceiling finishes was inaccurate because the maximum R- Square of comparison was less than 0.900. Therefore, the prediction was not at an acceptable level. However, the results of the questionnaire survey distinguished the significant design parameters for the construction cost of ceiling finishing.

Table IV-10-The level of impact for construction cost of ceiling finishes

	RII - Significance level of Impact	Mean	Level of significant	Rank
Ceiling area	0.891	3.563	Highly significant	1
Types of ceiling finishes	0.867	3.469	Highly significant	2
Numbers of floor	0.836	3.344	Highly significant	3
Gross internal floor area	0.664	2.656	Significant	4
Roof area	0.641	2.563	Significant	5

I. Impact on construction cost water supply and disposal system, HVAC and electrical installation

A building has different kinds of services. These services are commonly based on the function of the buildings. However, according to Yaman & Tas (2007), three main services have been listed water supply and disposal systems, heating, ventilation air conditioning and electrical insulation.

Table IV-11- Importance of building parameters on construction cost water supply and disposal system

	Importance	Normalized Importance	Rank
GIFA	0.589	100.0%	1
Total height of the building	0.145	24.7%	2
Numbers of floors	0.141	24.0%	3
Floor height	0.125	21.2%	4

The ANN model was unable to perform cost prediction of HVAC installation cost accurately because a limited number of projects had shown design details of HVAC systems in drawings initially. Because most of the low-rise residential projects had not decided on HVAC and cost it. Therefore, according to the results of a questionnaire survey, the significant level of parameters was as follows.

Table IV-12- The level of impact for the construction cost of Heating, ventilation, and Air conditioning

Rank	Design related building parameters	RII - Significance level of Impact	Mean	Level of significant
1	Numbers of floors	0.875	3.500	Highly significant
2	Volume of HVAC	0.859	3.438	Highly significant
3	Gross internal floor area	0.648	2.594	Significant
4	Floor height	0.648	2.594	Significant

The final main building element for this study is electrical installation. According to the BCIS (2012), electrical installation is defined as electric source and mains, power distribution, electric lighting distribution and fittings.

Table IV-13- Importance of building parameters on construction electrical installation

	Importance	Normalized Importance	Rank
GIFA	0.323	100.0%	1
Type of Electrical works	0.313	96.8%	2
Number of floors	0.203	62.8%	3
Total height of the building	0.120	37.2%	4
Internal wall area	0.026	8.1%	5
Floor height	0.014	4.3%	6

V. CONCLUSIONS

Cost prediction at the early stage is very important for successful project execution and completion. Parametric estimation is one of the accurate estimation methods that can be used in the conceptual stage of the project. However, elemental cost estimation is also an accurate estimation method. By considering that, this study develops the cost prediction model for the building element of the project.

First, the study proposed to consider the significant factors that have been listed under objective three before starting the project. They will be significant for successful project delivery. Under that, a study has identified that client-related factors are highly impacted on the construction cost of low-rise residential buildings. They are materials specified by the client, nature of the client, quality required by the client and financial ability of the client. Further, resource of material, labor and plant availability during the construction period also highly impacts construction costs. In addition to that, project duration, contract type and experience of the design team are highly significant factors. In addition, site location and quality of workmanship should be considered as high impact. Another nine factors are also categorised as significant for construction cost. However, these can be considered as general factors for any construction project.

This study focuses on low-rise residential buildings in Sri Lanka, which may limit the generalizability of the findings to other contexts or project types. Further research is needed to validate the model's applicability to different types of construction projects and geographical locations.

Further, this study has proposed elemental cost estimation based on the ANN model to prepare an accurate estimation. Based on the cost prediction, the importance of building parameters was measured. Therefore, this study recommends using building parameters which have obtained more than 10% of normalized importance for building elements. It will give a more accurate estimation at the conceptual stage. Based on this selected parameter, the cost estimation can be developed for each building element. Ultimately, all elemental costs can be used to develop an accurate conceptual estimation.

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